JalRakshak v2

AI-Powered Post-Flood Water Quality Monitoring & Disease Outbreak Prediction

JalRakshak

Team Northeastnerds

Built for Assam. Built for every flood-prone district in India.

AI | IoT | Disaster Response | PWA Offline

Problem Statement: Floods Trigger a Second Crisis

Assam floods kill people twice: first the water, then disease.

- Cholera, typhoid, hepatitis A and leptospirosis rise after floodwater contaminates wells and community sources.
- Communities lack real-time water safety intelligence and depend on delayed field sampling.
- Without early warning, contamination becomes a preventable public-health emergency.

5M+	40%	72h	0
displaced annually	deaths linked to disease	detection delay	real-time systems

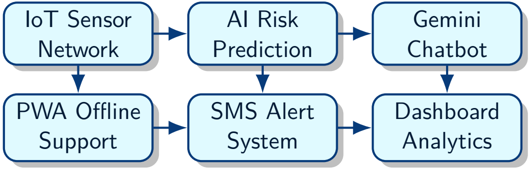
⚠ **No data, no prediction, no prevention**

Existing System Problems

- Manual reporting and field visits
- WhatsApp-based updates
- Delayed response and fragmented records
- No predictive analytics
- Limited multilingual support
- Internet failure during floods

Capability	Traditional System	JalRakshak
Detection	Delayed lab checks	Live pH, turbidity, TDS, temp
Prediction	Reactive	AI risk levels with confidence
Communication	Manual calls/messages	SMS, dashboard, chatbot
Access	Internet dependent	Offline-first PWA cache
Language	Mostly English/Hindi	English, Hindi, Assamese

JalRakshak: End-to-End Flood Water Intelligence

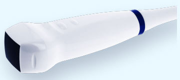


Core Idea

Transform raw floodwater readings into actionable disease-risk intelligence for citizens and health officials.

Hardware Overview: Flood Water Sensing Prototype

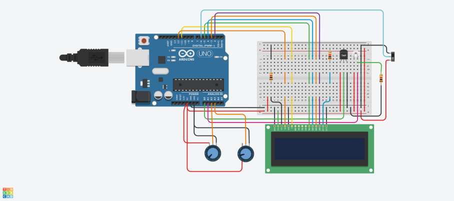
Water Probe Array



Arduino UNO
ADC + calibration

LCD Display
16x2 status

Future Cloud
API / MQTT

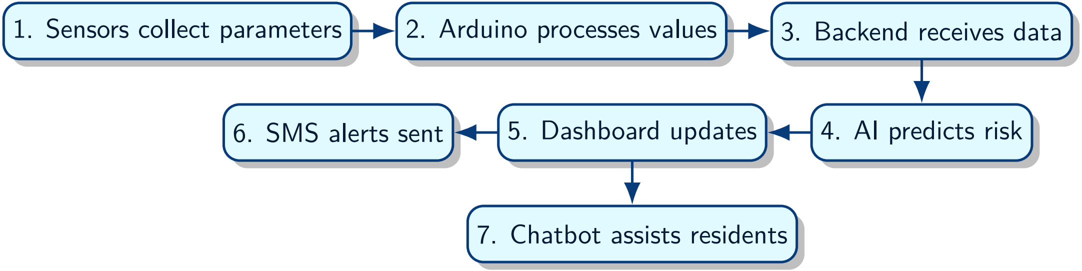


Prototype Note

This prototype continuously monitors water quality parameters during floods.

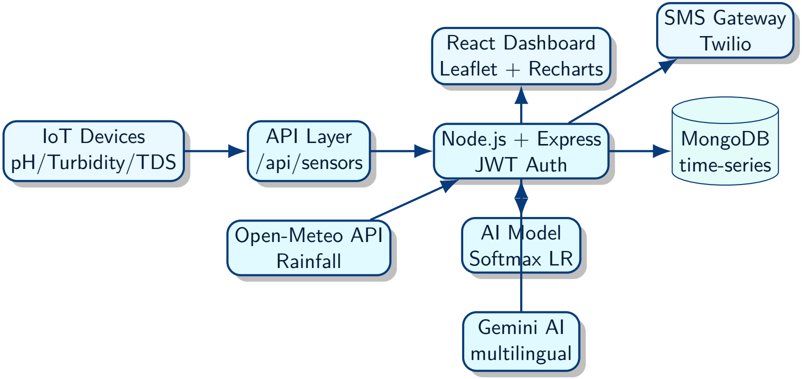
- Sensors collect pH, turbidity, dissolved solids and temperature.
- Arduino processes analog values using calibration constants.
- LCD shows real-time prototype readings.
- Breadboard, resistors and potentiometer stabilize sensor/LCD circuits.

Working Principle



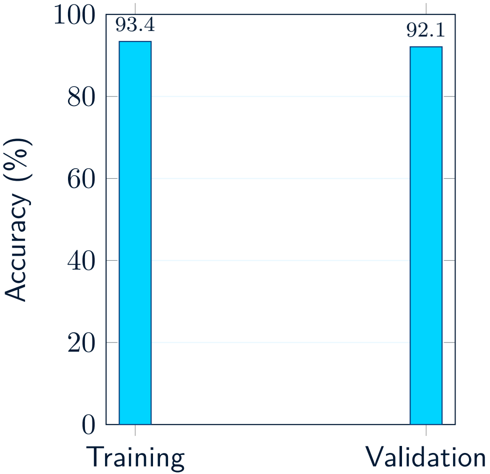
Transition-ready workflow for presentation mode

System Architecture



AI / Machine Learning Model

- Multinomial logistic regression with softmax classification.
- 12,480 CPCB-based Brahmaputra Basin water samples.
- Features: pH deviation, turbidity norm, contamination norm, rainfall norm, contamination $\times$ rainfall.
- Outputs: LOW MEDIUM HIGH CRITICAL



IoT & Real-Time Monitoring

- pH monitoring detects acidic/alkaline deviations.
- Turbidity and TDS indicate suspended solids and contamination risk.
- Rainfall from Open-Meteo improves outbreak context.
- Device metadata tracks GPS, signal strength, battery and calibration time.

Zone A
pH: 5.8 Turbidity: 72 NTU
RISK: Zone A

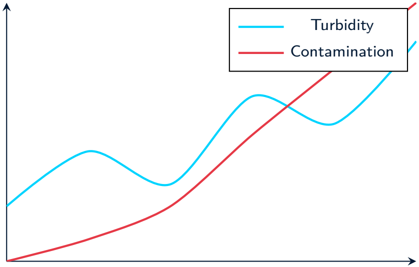
Zone B
pH: 6.1 Turbidity: 58 NTU
RISK: Zone B

Zone C
pH: 6.4 Turbidity: 44 NTU
RISK: Zone C

Zone D
pH: 7.1 Turbidity: 18 NTU
RISK: Zone D

Dashboard & Analytics

- Live zone dashboard with map visualization.
- MongoDB time-series storage with 7-day TTL.
- 24-hour pH, turbidity and contamination trends.
- Auto-refresh every 30 seconds with SSE alerts.



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AI Chatbot: Gemini-Powered Resident Assistant

- Multilingual support: English, Hindi and Assamese.
- Auto-detects language and replies in the same language.
- Injects live sensor context by zone.
- Provides safety guidance and emergency next steps.

Sample Conversation

Resident: Can I drink water in Zone A?

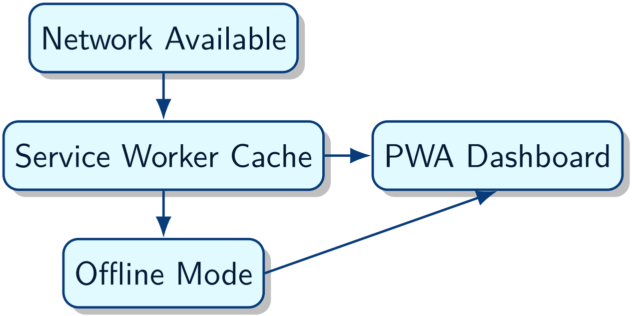
JalRakshak: Zone A is CRITICAL: pH 5.8, contamination 81%. Boil water or use bottled water until contamination drops below the safe threshold.

Hindi: Paani kab safe hoga?

AI: Jab risk LOW ho aur official alerts clear ho jayein.

Offline-First PWA

- Floods destroy connectivity when alerts matter most.
- Service workers cache app shell, map tiles, sensor snapshots and alerts.
- Network-first data with graceful fallback to last known readings.
- Offline banner keeps users informed.



Emergency Response Features

- SOS requests with location and status.
- Supply requests for water, food and medicines.
- Alert broadcasting by verified health officials.
- Emergency communication during blackouts.
- Public read-only access with official escalation.
- PDF reports for field teams and administrators.

Tech Stack

React

Node.js

Express

Leaflet

Recharts

Twilio

Arduino

Open-Meteo

Impact

- Faster outbreak detection and response.
- Reduced waterborne disease spread.
- Multilingual accessibility for local communities.
- Scalable deployment for flood-prone districts.
- Stronger disaster resilience and governance.

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Future Scope: Roadmap

- **Phase 1** — MQTT real devices
- **Phase 2** — Satellite flood mapping
- **Phase 3** — WHO integration
- **Phase 4** — Mobile app
- **Phase 5** — Nationwide deployment
- **Phase 6** — AI disease forecasting

Challenges Faced

- Sensor calibration across changing floodwater conditions.
- Network instability in submerged or remote areas.
- Data reliability and missing-value handling.
- Multilingual AI prompt design and safety.
- Offline synchronization after reconnection.
- Balancing low-cost hardware with trustworthy readings.

Demo / Prototype

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Demo Flow

Sensor reading → AI prediction → dashboard alert → SMS/chatbot guidance.

Conclusion

JalRakshak combines AI + IoT + Disaster Management to protect flood-prone communities.

- Converts water quality signals into real-time safety intelligence.
- Helps citizens avoid unsafe water before outbreaks spread.
- Gives officials a dashboard, alerts and reports for faster action.

From flood response to flood resilience.

Thank You

Questions?

Team Northeastnerds

Real ML | IoT Prototype | PWA Offline | AI Chatbot